

PHASE 8

FitBuddy- AI Fitness Plan Generator using Gemini Models

PROJECT DEMONSTRATION

1. Introduction

The Project Demonstration Phase is the final practical phase of the FitBuddy – AI Fitness Plan Generator project. The purpose of this phase is to demonstrate the working behavior of the developed application and explain how the different components work together.

FitBuddy is designed to generate personalized fitness plans using Generative AI. The application accepts user information such as name or user ID, age, weight, fitness goal, and workout intensity. Based on these inputs, the system generates a personalized seven-day workout plan and a concise nutrition or recovery tip.

The demonstration also explains how the application processes feedback. A user can provide feedback about the generated workout plan, and the system can use that feedback to create an updated version of the workout plan while preserving the original plan.

The project uses FastAPI for backend routing, Google Gemini for AI-powered generation, SQLite with SQLAlchemy for data persistence, and Jinja2 templates with HTML/CSS for the frontend.

The source project also documents an administrative view where stored users and their original or updated plans can be viewed.

2. Purpose of Project Demonstration

The main purpose of the demonstration is to show that the developed FitBuddy application can perform its intended workflow from beginning to end.

The demonstration focuses on:

- Starting the FitBuddy application.
- Opening the web interface.
- Entering user fitness information.
- Submitting the information to the backend.
- Generating a personalized workout plan.
- Displaying the generated workout plan.
- Generating a nutrition or recovery tip.
- Accepting user feedback.
- Updating the workout plan using feedback.
- Storing user and plan information.

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- Viewing stored users and plans.
- Explaining the AI, backend, database, and frontend integration.

The demonstration therefore connects the theoretical design and development work from the previous phases with the practical operation of the application.

3. Demonstration Objectives

The major objectives of the FitBuddy demonstration are:

Objective 1 – Demonstrate User Interaction

Show how a user interacts with the FitBuddy web application and provides the required fitness information.

Objective 2 – Demonstrate AI-Based Workout Generation

Show how the application uses the configured Generative AI service to generate a personalized workout plan.

Objective 3 – Demonstrate Nutrition Guidance

Show how FitBuddy generates a concise nutrition or recovery-related recommendation.

Objective 4 – Demonstrate Feedback Processing

Show how user feedback can be submitted and used to generate an updated workout plan.

Objective 5 – Demonstrate Data Persistence

Show how user information and generated plans are stored in the SQLite database.

Objective 6 – Demonstrate Backend Routing

Explain how FastAPI routes process requests and connect the frontend with application logic.

Objective 7 – Demonstrate the Complete System

Show the complete flow from user input to AI generation, database storage, feedback processing, and final display.

4. Demonstration Environment

The FitBuddy project is designed to run in a local development environment.

The documented technology stack includes:

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Component	Technology
Programming Language	Python
Backend Framework	FastAPI
AI Service	Google Gemini
Database	SQLite
ORM	SQLAlchemy
Frontend	HTML/CSS
Template Engine	Jinja2
Server	Uvicorn
Development Environment	VS Code

The project requires the required Python packages to be installed and the Google API key to be configured through the environment configuration.

The application can be accessed locally using:

`http://127.0.0.1:8000`

FastAPI also provides an interactive API documentation interface at:

`http://127.0.0.1:8000/docs`

The `/docs` interface can be used during the demonstration to explain the backend API and available routes.

5. Demonstration Preparation

Before beginning the demonstration, the following preparation should be completed.

Step 1 – Open the Project

Open the FitBuddy project folder in Visual Studio Code.

Step 2 – Check the Project Structure

Verify that the required source files, templates, services, database, and configuration files are available.

Step 3 – Install Dependencies

Install the Python dependencies listed in `requirements.txt`.

Step 4 – Configure the API Key

Configure the required Google API key in the environment configuration.

Step 5 – Start the Application

Run the FastAPI application using the project's configured Uvicorn command.

Step 6 – Open the Browser

Open:

`http://127.0.0.1:8000`

Step 7 – Verify the Application

Confirm that the FitBuddy homepage is displayed correctly before beginning the actual demonstration.

6. Demonstration Flow

The overall demonstration follows this sequence:

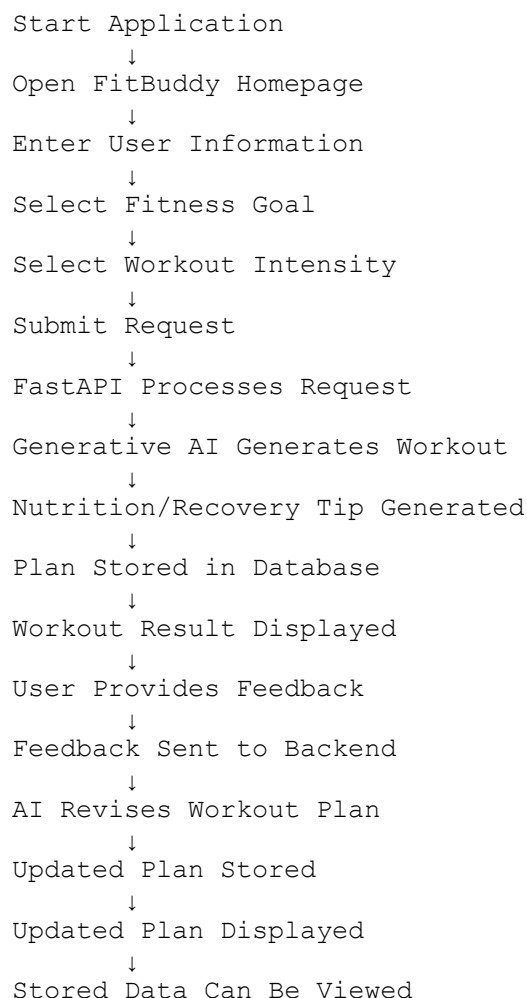


Figure 1: Complete FitBuddy Demonstration Flow

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This flow represents the sequence that can be explained during the final project demonstration.

7. Demonstration of the Homepage

The first screen demonstrated is the FitBuddy homepage.

The homepage acts as the starting point of the application. It provides the user with an interface through which fitness-related information can be entered.

The documented frontend uses HTML templates and CSS styling to provide a fitness-oriented interface.

The interface is designed to make the application easy to understand and use.

During the demonstration, the following points can be explained:

- Project name.
- Purpose of FitBuddy.
- User input section.
- Fitness goal selection.
- Workout intensity selection.
- Submit/generate action.
- Overall fitness-oriented visual design.

Figure 2: FitBuddy Homepage Demonstration

FITBUDDY

AI FITNESS PLAN GENERATOR

Name / User ID:

Age:

Weight:

Fitness Goal:

[Weight Loss]

Intensity:

[Moderate]

[GENERATE WORKOUT]

This figure is an illustrative representation of the demonstration interface and should not be treated as an actual screenshot.

8. Entering User Information

The next step is to demonstrate the user input process.

The user provides the information required by the FitBuddy application.

The documented input information includes:

- Name or user identifier.
- Age.
- Weight.
- Fitness goal.
- Workout intensity.

The information allows the application to create a personalized request rather than producing a generic workout plan.

For example, a demonstration can use sample information such as:

```
User ID: DemoUser
Age: 21
Weight: 65 kg
Fitness Goal: General Wellness
Workout Intensity: Moderate
```

These values are demonstration inputs and should be changed according to the actual demonstration requirements.

9. Fitness Goal Selection

FitBuddy supports different fitness objectives.

The documented goals include:

- Weight loss.
- Muscle gain.
- General wellness.

The selected goal becomes part of the information used for generating the workout plan.

For example:

```
Fitness Goal
├── Weight Loss
├── Muscle Gain
└── General Wellness
```

Figure 3: Fitness Goal Selection

During